

Multiple Choice Questions (MCQs) on Microservices in Java

1. **What is the primary characteristic of microservices architecture?**
 - A) Monolithic structure
 - B) Collection of small, independent services
 - C) Single database for all services
 - D) Centralized management
 - **Correct Answer:** B) Collection of small, independent services
2. **Which of the following is a benefit of microservices?**
 - A) Increased downtime
 - B) Scalability
 - C) Tight coupling
 - D) Single technology stack
 - **Correct Answer:** B) Scalability
3. **What does "loose coupling" in microservices mean?**
 - A) Services are tightly integrated
 - B) Changes in one service do not directly impact others
 - C) All services share the same database
 - D) Services are dependent on each other
 - **Correct Answer:** B) Changes in one service do not directly impact others
4. **Which framework is widely used for developing microservices in Java?**
 - A) Spring Boot
 - B) Angular
 - C) Django
 - D) Ruby on Rails
 - **Correct Answer:** A) Spring Boot
5. **What is the purpose of an API gateway in microservices architecture?**
 - A) To manage databases
 - B) To act as a single entry point for client requests
 - C) To handle service discovery
 - D) To enforce security policies
 - **Correct Answer:** B) To act as a single entry point for client requests
6. **Which of the following is NOT a deployment pattern for microservices?**
 - A) Multiple Service Instances per Host
 - B) Service Instance per Host
 - C) Monolithic Deployment
 - D) Serverless Deployment
 - **Correct Answer:** C) Monolithic Deployment
7. **What is a challenge associated with the fragmentation of business requirements in microservices?**
 - A) Increased performance
 - B) Data consistency
 - C) Simplified service ownership

- D) Enhanced fault tolerance
- **Correct Answer:** B) Data consistency

8. In microservices, what does "database per service" imply?

- A) All services share a single database
- B) Each microservice has its own database
- C) Databases are managed centrally
- D) No databases are used
- **Correct Answer:** B) Each microservice has its own database

9. Which of the following frameworks is used for building RESTful web services in Java?

- A) Spring Boot
- B) Dropwizard
- C) Jersey
- D) All of the above
- **Correct Answer:** D) All of the above

10. What is the main advantage of using the Blue-Green deployment pattern?

- A) Increased resource usage
- B) Simplified service discovery
- C) Minimizes downtime and allows for quick rollback
- D) Requires no additional resources
- **Correct Answer:** C) Minimizes downtime and allows for quick rollback

11. Which of the following is a key concept of microservices?

- A) Single point of failure
- B) Technology agnostic
- C) Centralized logging
- D) Synchronous communication only
- **Correct Answer:** B) Technology agnostic

12. What is the role of service discovery in microservices?

- A) To manage databases
- B) To automatically detect services on a network
- C) To enforce security policies
- D) To handle API requests
- **Correct Answer:** B) To automatically detect services on a network

13. Which of the following is a common challenge when implementing microservices?

- A) Simplicity in deployment
- B) Managing inter-service communication
- C) Reduced operational costs
- D) Increased data consistency
- **Correct Answer:** B) Managing inter-service communication

14. What is the purpose of using orchestration tools like Kubernetes in microservices?

- A) To manage databases
- B) To automate deployment and scaling of containers
- C) To enforce security policies
- D) To handle API requests

- **Correct Answer:** B) To automate deployment and scaling of containers

15. Which of the following is a feature of Spring Boot?

- A) Manual configuration only
- B) Auto-configuration
- C) No support for RESTful APIs
- D) Requires extensive boilerplate code
- **Correct Answer:** B) Auto-configuration